hdWGCNA Analysis Pipeline

Overview

This repository contains the analysis pipeline used for single-cell RNA sequencing (scRNA-seq) and high-dimensional weighted gene co-expression network analysis (hdWGCNA) performed in our submitted Cell Reports manuscript. The workflow integrates Seurat-based preprocessing with hdWGCNA network construction, module analysis, and hub gene identification.

Pipeline Summary

The pipeline includes:

- scRNA-seq preprocessing and quality control
- UMAP generation and cluster annotation
- Differential expression and marker identification
- Metacell construction
- hdWGCNA network analysis
- Soft-threshold optimization
- Module eigengene analysis
- Hub gene identification
- Network and module visualization

Software Requirements

R Environment
R ≥ 4.2 recommended

Required Packages
- Seurat
- hdWGCNA
- WGCNA
- tidyverse
- cowplot
- patchwork
- igraph
- SingleCellSignalR

Input Files

The workflow requires:

1. Integrated or processed Seurat objects (.rds)
2. Cluster annotation files
3. Optional custom gene lists for pseudobulk analyses

Workflow Summary

1. Seurat Preprocessing
Quality control visualization, filtering, normalization, scaling, PCA, UMAP generation, and cluster annotation.

2. Differential Expression Analysis
Cluster marker genes identified using Seurat-based differential expression analysis.

3. hdWGCNA Setup
Supports both variable-gene-based and pseudobulk/custom gene list analyses.

4. Network Construction
Includes soft-threshold power testing, TOM construction, eigengene computation, and connectivity analysis.

5. Hub Gene Analysis and Visualization
Generation of KME plots, network visualizations, and module UMAP embeddings.

Output Files

Major outputs include:

- Processed Seurat objects
- Marker gene tables
- Soft-threshold diagnostic plots
- TOM files
- hdWGCNA objects
- Hub gene tables
- Module and network visualizations

Notes

- Placeholder paths in the scripts should be replaced with local file paths prior to execution.
- Sample names, cluster IDs, and gene lists should be modified according to the dataset being analyzed.
- Random seed initialization was used for reproducibility.

Citation

If using this workflow, please cite:

- Langfelder and Horvath, WGCNA
- Morabito et al., hdWGCNA
- Hao et al., Seurat
- Saini et al., bioRxiv

Correspondence

For questions regarding the analysis workflow, please contact the corresponding authors of the associated Cell Reports submission.